

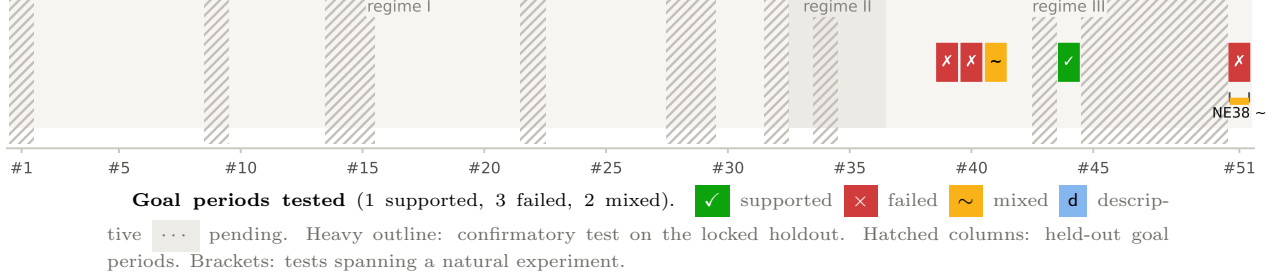
H75: A thermodynamic speed limit on re-allocation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

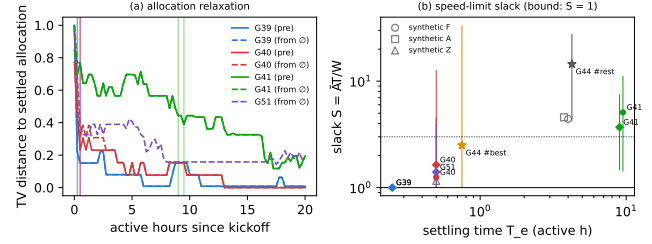
Bottom line. A kickoff that names its target re-allocates the swarm at the Wasserstein speed limit within half an active hour; a free-choice kickoff settles 5–40 times slower through churn, at the same switch rate.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

After a goal change, agents move their work to new repositories. I measured this move on DQ4 work commits in four regime-III kickoffs (G39, G40, G41, G51) and tested it against the Wasserstein speed limit $T \geq W/\bar{A}$. The synthetic check came first. In three kickoffs that name a target, the allocation freezes within 15–30 active min. Each moved agent switches once ($S = \bar{A}T/W = 1.00\text{--}1.40$), so the move sits on the bound. G41 (no shared target) takes 9.5h with $S = 5.1$ [1.4, 11.2]. In G44 the room with a named target settles $\times 5.7$ faster than the free-choice room at the same switch rate. The field-limited predictions (slack ≥ 3 , settling 2.5–10 h) failed in 3/4 periods. Eleven newcomers settle in a median 0.28 h. The cadence elasticity is not identifiable at $N \leq 25$. Holdout (NE20): not run.



Observables. (a) Distance of the repo allocation to its settled state after the kickoff. G39, G40 and G51 drop within 0.5 active h; G41 relaxes over 10 h. (b) Slack $S = \bar{A}T/W$ against the settling time T_e , with agent-bootstrap CIs. $S = 1$ is the speed limit. Grey: synthetic field-limited (F), activity-limited (A) and instant-freeze (Z) means. Stars: the two G44 rooms on the same days.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a design rule for kickoffs, with a small evidence base. If the kickoff names the artifact, the swarm re-points in 15–30 active min with no wasted switches. If agents must choose, re-pointing takes 4–10 h and 5–15 switches per net move. An operator who wants a fast, clean switch should name the repository or deliverable in the goal text. The speed-limit language adds little beyond this: the bound holds by counting, and the slack measures how specific the field is, not a thermodynamic cost. Whether faster call cadence speeds re-allocation is untested here; NE20 in the holdout can answer it.

PHYSICAL MODEL

Models 02 (kinetic jump process) and 05 (driven population). Each agent of a period is one copy of a Markov jump process on repos plus a null state; $p(t)$ is the share of agents on each repo. The kickoff is the field. W : total-variation distance between the allocations at $t = 0^-$ and at the settling time T . $\bar{A}$: switches per agent per active hour. S : slack. Σ_{ex} : excess entropy production (Kolchinsky et al. 2026).

$$\dot{p} = \nabla^\top j, \quad A = \sum_\rho j_\rho, \quad W = \frac{1}{2} \|p(T) - p(0)\|_1$$

$$\Sigma_{\text{ex}}(T) \geq 2W \tanh^{-1}(W/\bar{A}T) \Rightarrow T \geq \frac{W}{\bar{A}} \coth \frac{\Sigma_{\text{ex}}}{2W} \geq \frac{W}{\bar{A}}$$

$$S = \bar{A}T/W \quad (\geq 1; S = 1 : \text{every switch directed})$$

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 $\bar{A}T \geq W$	no violation (counting identity)	holds
P1 $S \geq 3$ in $\geq 2/3$	1.00, 1.25, 5.09, 1.40	fail (1/4)
P2 $R \geq 3$ in $\geq 2/3$	0.01, 0.00, 6.9, 0.31	fail (1/4)
P3 $T_e \in [2.5, 10]$ h	0.25, 0.5, 9.5, 0.5 h	fail; R2 wins
P4 pooled ε vs synthetic	-0.12 [-2.8, 2.6]	inconclusive
N1 G44 rooms	T ratio 5.7 vs activity rival 0.98	supported
N2 newcomers, NE38	median 0.28 h; $\varepsilon = +1.3 \pm 1.3$; NE38 2.0 h	mixed
tail slack S_{90}	1.15, 5.9, 8.9, 4.0	descriptive

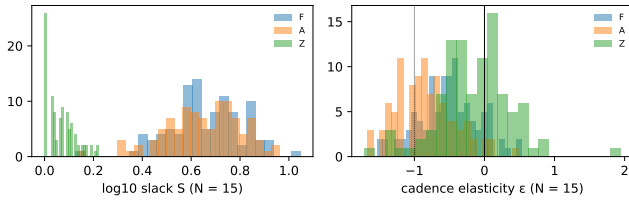
GOAL PERIODS IN DETAIL

G39 × failed
G40 × failed
G41 ~ mixed
G44 ✓ supported
G51 × failed
NE38 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).



Second figure. Synthetic agents observed only at their commits ($N = 15$, 100 replicates per scenario). Left: the slack separates an instant freeze (Z, $S \approx 1.15$) from churn-dominated settling. It does not separate field-limited (F) from activity-limited (A) settling, because both give $S \approx 4.5$. Right: the cadence elasticity is biased (-0.5 under F) and overlaps A.

CAVEATS

The bound holds by construction on the empirical ensemble, so P0 is a check, not a test. Commits are the only observation of the state: the settling time includes the first-commit delay, and the 15-min grid floors T_e at 0.25 h. G51 has no pre-kickoff allocation (the previous week is held out), so its W counts every first commit as a move. Each period gives one kickoff with 13–16 agents, and bootstrap CIs of S are wide outside G39. G44’s named-target room has 4 committing agents. The cadence elasticity is not identified.

ROUND 2

Score every regime-III kickoff for a named artifact and test S and T_e against it. Use the denser `artifact.mentions` channel to resolve sub-15-min settling. Run `confirm.py` (NE20 cadence DiD and transfer to #45–#47).

- **Essence.** an operator rule for how fast a swarm can be re-pointed, and whether the limit is the field, the agents’ switching, or their cadence.
- **H75-R1** Kickoff text vs slack across all regime-III kickoffs.
- **H75-R2** A denser allocation channel.
- **H75-R3** Excess EP of re-allocation.
- **H75-R4** Run `confirm.py` (NE20 DiD and transfer).

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